Definition 8.6

Two or more vectors are said to be **collinear** or **parallel** if they have same line of action or have the lines of action parallel to one another.

Two or more vectors are said to be **coplanar** if they lie on the same plane or parallel to the same plane.

Definition 8.7

Two vectors are said to be **equal** if they have same direction and same magnitude.

Let us note that it is not necessary to have the same initial point and same terminal point for two equal vectors. For instance, in Fig. 8.2, the vectors $\vec{b}$ and $\vec{c}$ are equal since they have same direction and same length, whereas $\vec{a}$ and $\vec{b}$ are not equal because of opposite direction even though they are having same length. The vectors $\vec{c}$ and $\vec{d}$ are not equal even though they are having same direction but not having same length.

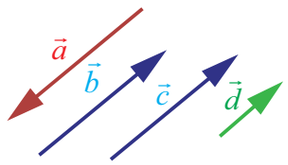


Fig. 8.2

Definition 8.8

Zero vector is a vector which has zero magnitude and an arbitrary direction and it is denoted by $\vec{0}$.

That is, a vector whose initial and terminal points are coincident is called a zero vector.

We observe that the initial and terminal points of a zero vector are the same. The zero vector is also called **null vector** or **void vector**.

A vector of magnitude 1 is called a **unit vector**. The unit vector in the direction of $\vec{a}$ is denoted by $\hat{a}$ (read as 'a cap' or 'a hat'). Clearly $|\hat{a}| = 1$.

We observe that there are infinitely many directions and hence there are infinitely many unit vectors. In fact, for each direction there is one unit vector in that direction.

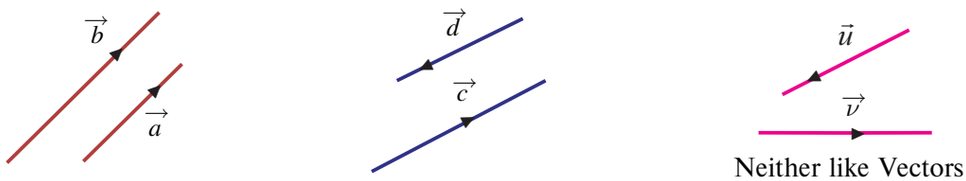
Any non-zero vector $\vec{a}$ can be written as the scalar multiple of a unit vector in the direction of $\vec{a}$. This scalar is nothing but the magnitude of the vector.

Thus for any vector $\vec{a} = |\vec{a}|\,\hat{a}$, where $\hat{a}$ is the unit vector along the direction of $\vec{a}$.

Clearly $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$ for any non-zero vector.

Definition 8.9

Two vectors are said to be **like vectors** if they have the same direction. Two vectors are said to be **unlike vectors** if they have opposite directions.



Like Vectors | Unlike Vectors | Neither like Vectors nor unlike Vectors

Fig. 8.3

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